Set Up a Web Application on Amazon EC2

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1. Objective

To launch a virtual server on Amazon EC2, secure it with a Security Group that allows SSH (port 22) only from my IP and HTTP (port 80) for public access, connect to it over SSH, install the Nginx web server, deploy a simple HTML web application, and verify it in a web browser.

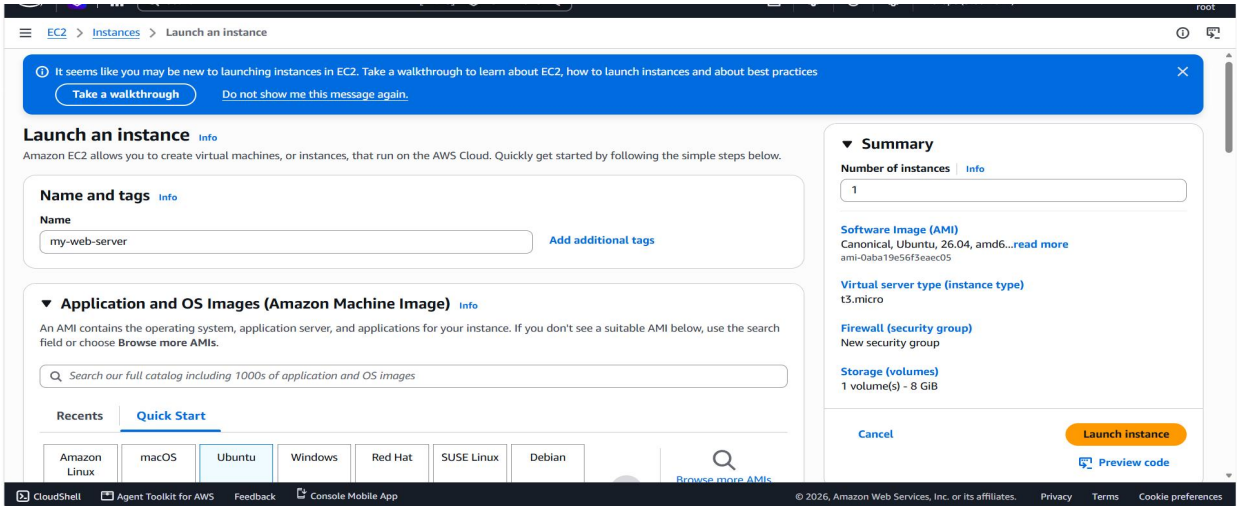
2. Architecture

User Browser → Internet → Security Group (ports 22, 80) → EC2 Instance (Ubuntu) → Nginx → index.html

Component	Role
EC2 instance	Virtual server hosting the application
Key pair (.pem)	Private key for secure SSH authentication
Security Group	Virtual firewall controlling inbound and outbound traffic
Nginx	Web server listening on port 80
index.html	The deployed web application

3. Step 1: Launch an EC2 Instance

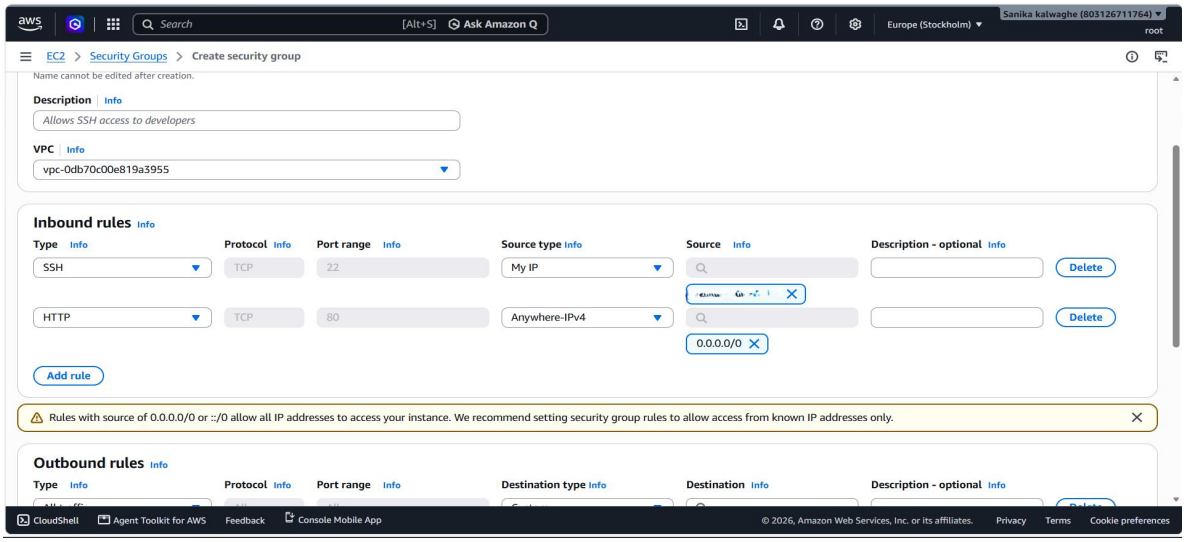
1. In to the AWS Management Console, opened EC2, and clicked Launch instance.
2. Name: my-web-server
3. AMI: Ubuntu Server 24.04 LTS (Free tier eligible)
4. Instance type: t2.micro / t3.micro
5. Key pair: created my-ec2-key (RSA, .pem format) and stored it safely.
6. Storage: default 8 GB gp3.
7. Clicked Launch instance.



Step 2: Configure the Security Group

A security group named web-server-sg was configured with these inbound rules:

Type	Protocol	Port	Source	Purpose
SSH	TCP	22	My IP	Admin access only from my machine
HTTP	TCP	80	0.0.0.0/0	Public web access



Step 3: Connect to the Instance via SSH

Command run from Windows PowerShell:
ssh -i "C:\Users\Asus\Downloads\my-ec2-key.pem" ubuntu@ip

Step 4: Install and Configure Nginx

sudo apt update && sudo apt upgrade -y
sudo apt install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
sudo systemctl status nginx

Step 5: Deploy a Simple HTML Web Application

cd /var/www/html
sudo mv index.nginx-debian.html index.nginx-debian.html.bak
sudo nano index.html

Contents of index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My EC2 Web App</title>
  <style>
    body { font-family: Arial, sans-serif; background: #f0f4f8; text-align: center;
padding-top: 80px; }
    h1 { color: #232f3e; }
    p { color: #555; }
    .badge { display: inline-block; background: #ff9900; color: #fff; padding: 8px 16px;
border-radius: 6px; }
  </style>
</head>
<body>
  <h1>Hello from Amazon EC2!</h1>
  <p>This web application is hosted on an Ubuntu EC2 instance using Nginx.</p>
  <span class="badge">Deployed by: YOUR NAME</span>
</body>
</html>
```

```
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Oct 18 04:58:32 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   30.5% of 6.61GB   Processes:             116
  Memory usage: 25%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.33.135

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-33-135:~$
```

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Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

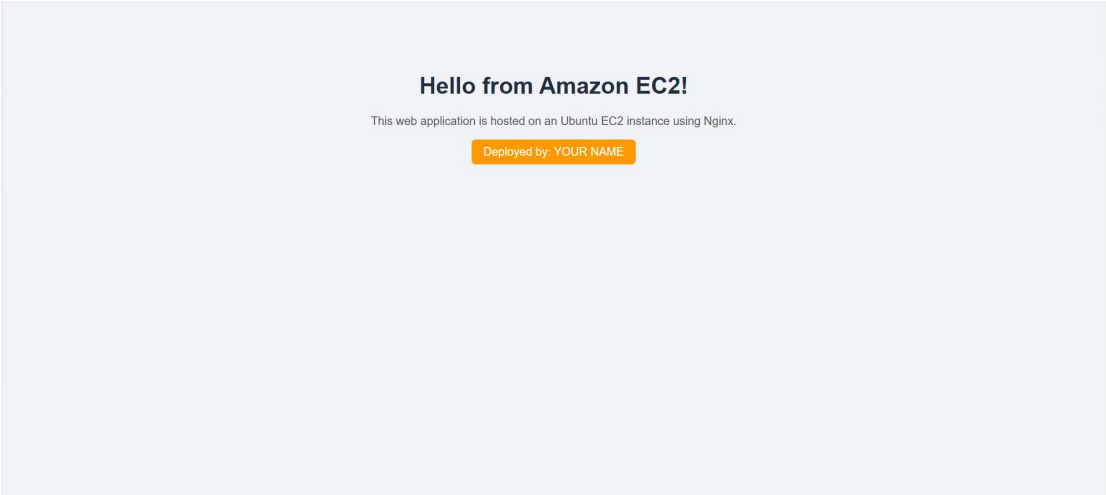
Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart getty@tty1.service
 systemctl restart networkd-dispatcher.service
 systemctl restart serial-getty@ttyS0.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 ubuntu @ session #1: sshd-session[1159,1276]
 ubuntu @ user manager: (sd-pam)[1166]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-33-135:~$ sudo systemctl start nginx
ubuntu@ip-172-31-33-135:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-172-31-33-135:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-10-18 05:04:26 UTC; 36s ago
 Invocation: 4f63030c5ef74769a61bdd0fa653f7db
       Docs: man:nginx(8)
   Main PID: 28353 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3.1M (peak: 7.3M)
        CPU: 42ms
     CGroup: /system.slice/nginx.service
             ├─28353 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─28356 "nginx: worker process"
             └─28357 "nginx: worker process"

Oct 18 05:04:26 ip-172-31-33-135 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 18 05:04:26 ip-172-31-33-135 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-33-135:~$
```



Conclusion

An EC2 instance was launched, secured with a least-privilege Security Group, accessed over key-based SSH, configured with the Nginx web server, and used to host a simple HTML application reachable from a browser. The exercise covered virtual

servers, network security, remote administration and web deployment, along with practical troubleshooting of connectivity problems.